



Faculty of Health Sciences

Department of Optometry

Clinical and Community
Practice Manual
version 4

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Preface by the Head of Department

Dear Students,

As the Head of the Department, it is with great pride that I present to you this latest edition of our Clinical Community Manual. This manual represents the collective knowledge and dedication of our esteemed faculty and experts in the field of optometry and ophthalmology. Our goal has always been to provide a resource that is not only rich in content but also practical in application.

In this edition, we have incorporated the latest advancements in eye care, ensuring that you are well-equipped to meet the challenges of modern clinical practice. We understand the dynamic nature of healthcare and the importance of staying current with emerging techniques and technologies. This manual reflects our commitment to excellence in education and patient care.

I would like to extend my heartfelt gratitude to all contributors for their invaluable insights and tireless efforts in bringing this manual to fruition.

Dr. Ogbonna Grace: For her expertise and invaluable insights in the field of optometry, which greatly enriched the content of this manual.

Dr. Timothy Chikasirimobi: Whose knowledge and dedication to ophthalmology have been instrumental in shaping the clinical guidelines detailed within these pages.

Mr. Kanayika Emmanuel: For his technical expertise and commitment to ensuring that the manual reflects the latest advancements and practices in eye care.

Mr. Kabwaza Innocent: Whose meticulous attention to detail and profound understanding of clinical procedures have significantly contributed to the comprehensiveness of this manual.

Mr. George Moyo: For his extensive contributions, particularly in the areas of curriculum design and the integration of innovative teaching methodologies.

Their collective efforts have been pivotal in creating a resource that is both educational and practical, serving the needs of students and practitioners alike. We are deeply appreciative of their dedication and commitment to the advancement of eye care education and practice.

To our students and fellow practitioners, I hope this manual serves as a beacon of knowledge, guiding you towards a successful and fulfilling career in eye care.

May this manual inspire you to continue learning, growing, and contributing to the health and well-being of our communities.

Sincerely,
Afonne C Joseph
Head of Department
Optometry

Declaration

I,..... hereby solemnly affirm my commitment to adhere to the rules, regulations, and established procedures as delineated in the Mzuzu University Student Clinical Manual. I acknowledge and understand that the clinical courses encompass both preclinical practice, which involves working with student partners, and clinical practice, which entails providing care to actual patients. Additionally, I accept full responsibility for any equipment or other items issued to me by the department for the purpose of practice and patient examination in a clinical setting. I also understand and agree that in the event of damage or loss due to theft or any other cause, I am liable to reimburse the university for the cost of replacing the item.

Student signature..... Date.....

1.0 INTRODUCTION

This student manual has been compiled as a guide and resource reference for clinical students studying optometry. It comprises competencies, objectives, procedures, rules, and forms. The students should therefore acquaint themselves with all the information contained in this manual.

1.1 COURSE COMPETENCIES AND OBJECTIVES

1.1.1 Competencies

Level three

- Complete a full and accurate refractive and ocular health examination within 60 minutes.
- Complete a comprehensive dilated eye exam in 2 hours.
- Complete a targeted case history
- Complete an accurate retinoscopy and refraction in a maximum of 30 minutes controlling the patient's accommodation.
- Offer an accurate list of differential diagnoses and suggest a logical clinical plan to include supplementary testing, dilation, cycloplegia, etc.
- Complete an accurate internal and external ocular health assessment. Attention to physical and clinical observation will be emphasized.
- Perform Goldman tonometry without abrading the cornea extensively.
- Understand how to manage most ocular pathologies and refractive conditions.
- Educate patients using appropriate communication and layman's terms.
- Suggest appropriate additional diagnostic tests.
- Make proper surgical medical referrals.
- Identify emergent and urgent clinical situations and the appropriate management.
- Record accurate and concise assessment and plan using sound clinical judgment.
- Seek out relevant research to support the latest treatment and management of most conditions
- Students should be able to examine in a manner that respects the professional and ethical boundaries of the profession of optometry.

Level four

- Complete a full refractive examination including ocular health assessment within 60 minutes.
- Complete a comprehensive dilated examination within 90 minutes
- Manage most clinical scenarios with minimal assistance.
- Manage difficult patients appropriately with minimal assistance.
- Initiate further appropriate testing if the patient's symptoms or examination results suggest the need.
- Seek out relevant research to support the latest treatment and management of most conditions.
- Students should be able to examine in a manner that respects the professional and ethical boundaries of the profession of optometry.

Level five

- Complete a full refractive examination including ocular health assessment within 45 minutes.
- Complete a comprehensive dilated examination within 90 minutes.
- Manage most clinical scenarios with minimal assistance.
- Manage contact lens, low vision, binocular vision, pediatric and pre, and post-operative surgical patients.
- Manage difficult patients appropriately with minimal assistance.
- Initiate further appropriate testing if the patient's symptoms or examination results suggest the need.
- Have a working knowledge of relevant research to support the treatment and management of most conditions.
- Communicate appropriately with other health care providers regarding patient management, consultations, and referrals.
- Students should be able to examine in a manner that respects the professional and ethical boundaries of the profession of optometry.

1.1.3 Objectives

Upon successful completion of the pre-clinical courses, students will acquire the competencies to execute the following tasks with proficiency within the clinical setting:

1. Conduct Comprehensive Patient Case Histories:

Demonstrating the ability to elicit detailed and relevant information about the patient's medical and ocular history.

2. Perform Problem-Oriented Ocular Assessments:

Skillfully conducting assessments focused on the patient's specific ocular issues.

3. Execute Internal and External Ocular Health Assessments:
Systematically carrying out evaluations of both internal and external ocular health.
 4. Ascertain Patients' Refractive Errors:
Accurately determining the nature and extent of refractive errors in patients.
 5. Engage in Differential Diagnosis:
Utilizing critical analysis of collected data to engage in differential diagnosis, leading to informed and accurate final diagnoses.
 6. Manage Ophthalmic Prescriptions:
Competently handling prescriptions related to ocular conditions.
 7. Administer Management of Common Anterior Segment Diseases:
Effectively managing common diseases of the anterior segment of the eye.
 8. Provide Appropriate Patient Counseling:
Offering relevant and empathetic counseling to patients regarding their ocular conditions and management plans.
 9. Refer Patients to Supplementary Care Levels When Necessary:
Making appropriate referrals to other healthcare professionals or advanced care levels as required by the patient's condition.
 10. Maintain Accurate and Complete Patient Records:
Diligently documenting and preserving comprehensive patient records, ensuring accuracy and confidentiality.
- These objectives ensure that students are thoroughly prepared for practical, patient-centered care in the field of optometry.

1.2 CLINICAL COURSE DESCRIPTION

The courses are intended to provide students with supervised clinical practice experience to meet requirements and help them acquire the necessary skills.

	LEVEL	COURSE CODE	COURSE TITLE	CREDIT
3	Semester 5			

		BSOP 3504	Clinical and community practice IA	8
	Semester 6			
		BSOP 3603	Clinical and community practice IB	8
4	Semester 7			
		BSOPT 4704	Clinical and community practice IIA	15
	Semester 8			
		BSOPT 4805	Clinical and community practice IIB	15
5	Semester 9			
		BSOPT 5104	Clinical and community practice IIIA	44
	Semester 10			
		BSOPT 5205	Clinical and community practice IIIB	44

1.3 CLINICAL COURSE POLICIES AND PROCEDURES

Attendance

- Students enrolled in the Bachelor of Science in Optometry (Honors) program are obligated to attend all face-to-face and virtual classes. If a student misses a clinical course or clinic, they must notify the clinical course instructor. Students must notify the preceptor and clinical coordinator before the scheduled start time for all emergencies or absences from the clinical site.

Professionalism attire

Students are expected to maintain high standards of personal cleanliness. It is imperative that they wear clean lab coats and attire at all times.

The dress code for clinical settings should be conservative and modest, adhering to the following guidelines:

- Shirts: Long or short-sleeved button-down shirts, accompanied by a tie, are considered appropriate.
- Skirts/Dresses: The length of skirts or dresses should not exceed 2 inches above the knee to ensure modesty.
- Jeans: Wearing jeans is deemed inappropriate for clinical settings.
- Footwear: Students must wear clean, formal school shoes. Sports shoes, as well as shoes with open toes or open heels, are not permissible in clinical areas.

These guidelines are established to promote a professional environment conducive to clinical learning and patient interaction.

Hair

Regarding personal grooming and appearance, students are required to adhere to the following standards:

- Hair for Female Students: Hair should be kept neat and securely tied back using suitable hair bands to maintain a professional appearance.
- Hair and Facial Hair for Male Students: Beards and mustaches must be kept clean and neatly trimmed.
- Hair Color: Students should avoid hair colors that are noticeably unnatural or that could be distracting in a clinical setting.
- Tattoos: Any tattoos should be covered while at the clinical site to ensure a professional and uniform appearance.

Nails

In maintaining a professional and hygienic appearance, students are required to adhere to the following guidelines regarding nail care:

- Artificial Nails: The use of artificial nails is strictly prohibited.
- Natural Nail Care: Nails should be kept clean and trimmed short.

1.4 Clinical Supervision Procedures

- Levels 4 and 5 of the program are structured to alternate between clinical and classroom learning, each phase lasting a specified number of weeks, such as 5 weeks in the clinic followed by 5 weeks in class.
 - This rotation is designed so that while fifth-year students are in class, fourth-year students are engaged in clinical practice.
- Upon arriving at the clinical site, the designated group leader is responsible for notifying the person in charge of their arrival.
 - Throughout their clinical rotation, students are required to be present consistently at the clinic as per the schedule, unless alternative arrangements are made with the clinic supervisor.
- Students are expected to adhere to the roster set by the Clinical Supervisor for topic discussions and case presentations.

- These presentations may occur either before or after the clinic hours on any given day. It is mandatory for each student to present at least one case and one topic during their time at each clinical site.
- The student group is scheduled to depart for the clinic on Saturday or Sunday and return to campus immediately after their assessment, marking the completion of their clinical placement for that week.
- At the conclusion of each clinical placement, students are responsible for ensuring that all patient forms are submitted to the Clinical Supervisor.
 - Additionally, they must verify that patient cases have been thoroughly graded and recorded in both their own assessment forms and those of the supervisor.
 - This step is crucial to maintain accurate and comprehensive records of the clinical experience.

1.5 Patient/Case Expectations per Semester

For each academic level in the optometry program, students are required to meet specific patient encounter quotas to ensure a comprehensive and diverse clinical experience. These requirements are outlined as follows:

Level 3 Requirements:

Each student must attend to a total of 40 patients throughout the academic year.

The cases must include a range of conditions:

- Dry Eyes: 2 cases
- Corneal Ulcers: 2 cases
- Myopia: 4 cases
- Hyperopia: 4 cases
- Astigmatism: 4 cases
- Ocular Trauma: 4 cases
- Bacterial Conjunctivitis: 2 cases
- Cataract: 2 cases
- Viral Conjunctivitis: 4 cases
- Uveitis: 4 cases
- Glaucoma: 4 cases
- Pinguecula: 2 cases
- Pterygium: 2 cases

Level 4 Requirements:

Each student is required to see 50 patients over the academic year.

This quota includes all cases required for Level 3, plus an additional 10 cases covering:

- Hypertensive Retinopathy
- Diabetic Retinopathy
- Glaucoma
- Congenital Cataract
- Hyphema
- Posterior Vitreous Detachment
- Uveitis
- Presbyopia
- Dry Eye
- Foreign Body on the Cornea

Level 5 Requirements:

Students must manage 60 patients throughout the year.

This includes all cases from Level 3, plus an additional 20 cases covering at least one each of:

- Hypertensive Retinopathy
- Diabetic Retinopathy
- Glaucoma
- Senile and Congenital Cataract
- Hyphema
- Posterior Vitreous Detachment
- Presbyopia
- Corneal Ulcers
- Ocular Trauma
- Bacterial Conjunctivitis
- Viral Conjunctivitis

Note: It is the responsibility of the students to ensure they meet these patient/case quotas each academic year. The grades for Clinical and Community practice will not be finalized for the second semester if these requirements are not met, unless a waiver is granted for valid reasons. This policy is in place to ensure that students gain adequate exposure to a wide range of clinical scenarios, essential for their professional development and competency in the field of optometry.

2.0 Clinical Examinations

Clinical examination for each level will be formulated to cover the objectives for each respective level of study for each semester.

2.1 Clinical Examination for each level

The Clinical Examination for each semester should cover the following:

Level 3

- Case history taking
- Preliminaries
- Tonometry
- Slit lamp
- Refraction
- Basic binocular vision assessment
- Fundoscopy-direct and indirect
- Visual fields
- Diagnosis
- Management of ocular diseases

Level 4

- Case history taking
- Preliminaries
- Tonometry
- Slit lamp
- Binocular vision assessment
- Refraction
- Fundoscopy-direct and indirect
- Diagnosis
- Management of ocular diseases

Level 5

- Case history taking
- Preliminaries
- Tonometry
- Binocular vision assessment
- Gonioscopy
- Pachymetry
- Visual field testing and interpretation
- Pre-contact lens assessment
- Contact lens insertion and removal
- Slit lamp
- Refraction
- Fundoscopy-direct and indirect
- Advanced testing
- Diagnosis
- Management of ocular diseases

2.2 Examination Procedure

The Clinical Assessment for each semester will be held in the last week of the clinical placement. The department, in consultation with the preceptor, will set the date for the exam. The clinical supervisor/preceptor, along with an academic staff member from Mzuni, will conduct the exam at the clinical area. The clinical examination will include a full eye examination. The grades from this exam will account for 40% of Clinical and Community Practice for each semester, while Topic and Case Presentation will account for 10% and Clinical Encounter Grades will account for 50%.

2.3 Log Book and Clinical report

STEPS FOR USE OF THE LOG BOOK AND ATTACHMENT REPORT WRITING by the Students

The logbook for optometry students is a tool that allows them to conduct a well-documented and comprehensive clinical evaluation during their attachment. The logbook records all activities performed by the student, the doctor/intern/supervisor during the attachment period, and also allows the student to comprehensively and carefully perform and report all clinical activities.

The log book document is in 3 parts:

- Part 1: Guidelines for writing and submission of clinical attachment report
- Part 2: Clinical attachment assessment Form for assessment by the Supervisor from Mzuni, Hospital/clinic supervisor/in-charge and the student.
- Part 3: Clinical attachment activities Log book

Step 1: Preparations

- Print, arrange and NEATLY bind (Tape binding is preferable) the entire document
- Take the document for endorsement by the attachment coordinator
- Read, familiarize, assimilate and master the guidelines and instruction on filling the document.

Step 2: Filling the attachment log book

- Ensure that the Doctor/clinic supervisor/examiner duly fill the section “Feedback on students’ performance”.
- Ensure that as a student, you duly, truthfully and transparently fill the section “Students feedback form”.
- Ensure that the Doctor/Clinical Supervisor duly fill, signs and put the establishment stamp on the section “Clinical Attachment Clearance Form”.
- Observe the stated guidelines for marking/scoring “Guidelines for marking clinical

- attachment report”. (For the Doctor/Clinic supervisor/examiner Only).
- Ensure that the Doctor/Clinic supervisor/examiner duly fills as demanded, tick as
- required and Record as provided and compute the scores in the “Summary of Scores” section.

Please note that your attachment grade/score/mark will be partly determined by this section. The minimum grade/score/mark is 0, and the maximum grade/score/mark is 65.

Step 3: Attachment report writing

- In accordance with the Daily/end of Week attachment activities, write your daily clinical activities report and also the reflection of the day and end of week – clinical attachment Activities summary/Reflection in a complete, clear, transparent, and ethical manner.
- Ensure that both Doctor/clinic supervisor/examiner and students signs the log book, daily and weekly as provided (Part 3: Clinical attachment activities log book)
- Ensure that you compile your attachment report as instructed (Daily, daily reflections, Weekly. See Part 1: Guidelines for writing and submission of clinical attachment report of the attachment booklet.
- Add extra paper if the provided space is not enough while writing the attachment daily and end of week report/reflections.
- Reports written separately, should be **NEATLY** bound (Tape binding is preferable).
- Submit your attachment report not more than one week from the end of the attachment day as directed.
- Document should be submitted through the secretary of the department of Optometry, Faculty of health sciences. **The task must be completed within one week of the end of the attached document.**

Please note that your attachment's grade/score/mark is partially based on the following section: Report. The grade/score/mark for this section is 100. Please see the score sheet in the last section of part 1 of the document for more information.

3.0 CLINICAL PROTOCOLS

3.1 CASE HISTORY

3.1.1 Chief Complaint

- Ask questions to specifically elicit the presenting complaint
- Ask about the onset of presenting complaint
- Ask about the duration of presenting the complaint
- Ask about the frequency of presenting complaint
- Ask about the laterality of presenting a complaint
- Ask about the location of presenting the complaint
- Ask about the pain of presenting a complaint
- Ask about the progression of presenting a complaint
- Ask about the severity of presenting a complaint
- Ask about the relief of presenting complaint
- Ask about associated signs and symptoms of presenting complaint
- Ask about vision specifically

Ocular history

- Ask about the last eye exam. (if yes, when, where, and outcome)
- Ask about the history of ocular disease/trauma
- Ask about the history of ocular surgery
- Ask about the history of spectacles

Medical history

- Ask about previous or current health problems (DM, HTN, etc.)
- Ask about the last medical exam (when, where & outcome)
- Ask about current medical drug use

Family ocular and medical history

- Ask about the history of family ocular conditions
- Ask about a family history of low vision and/or blindness
- Ask about spectacle use
- Ask about the history of medical conditions.

3.1.2 VISUAL ACUITY

- Explain the purpose and procedure of the test to the patient.
- Instruct the patient to cover up the left eye with the palm of the left eye.

- Ask the patient to read letters from the top big letters to the smallest letters that the patient can manage to read.
- Then cover up the right eye in the same way the left eye was occluded and ask the patient for the letters from the top letters to the smallest on the chart.
- Lastly, measure the VA with both eyes open.
- Record the VA for OD, OS, and OU.

Pinhole

If the patient's VA is 6/18 and worse then use the pinhole to measure the VA.

Put the pinhole in front of the right eye while the left eye is occluded.

- Instruct the patient to look through the hole to read the letters on the chart.
- Measure the VA for OD and then OS.
- Record the Pinhole VA for each eye.

Finger counting

- If the patient cannot read the letters on the chart even when the test distance has been reduced to 1 meter, then do the finger counting test.
- Explain to the patient the procedure.
- Instruct the patient to occlude one eye at a time, LE for instance.
- Raise fingers in front of the patient at a distance of 20 cm and ask the patient to count the number of fingers.
- Then do the same to the LE when the RE is occluded.
- Record the VA as OD: FC@ 20cm or OS: FC@20cm.

Hand movement

- If the patient cannot count fingers at 20cm then proceed to measure VA using hand movement.
- Explain to the patient the procedure.
- Instruct the patient to occlude one eye at a time, LE for instance.
- Raise the palm and wave it in front of the patient at a distance of 20 cm and ask the patient to say what he/she sees.
- Then do the same to the LE when the RE is occluded.
- Record the VA as OD: HM@ 20cm or OS: HM@20cm.

Light projection/perception

- If the patient cannot count fingers/ or see the hand presented, then proceed to a light projection test.
- Explain to the patient the procedure.
- Instruct the patient to occlude one eye at a time, LE for instance.
- Shine the light in all quadrants and ask what he/she sees.

- Then do the same to the LE when RE is occluded.
- Record the VA as OD: LP or OS: NLP.

3.1.3 COVER TEST

- Wash hands and disinfect equipment
- Introduce yourself and the purpose of the procedure avoiding the use of jargon
- Sit directly in front of the patient about 25 to 40 cm from the patient
- Ensure the appropriate room lighting. (Standard room illumination)
- Provide an appropriate target for the patient. (Consider distance and near target)
- Choose the correct target one line above poorer VA, preferably letters at either end of the line
- Observe fixation stability for 5 to 10 seconds
- Remind patient to fixate on target throughout the procedure
- Perform procedure stating the type of cover test
- Ensure the correct distance between Occulder and eye
- Introduce prism to estimate the magnitude of deviation
- Record using standard notation
- Perform procedure smoothly

3.1.4 AMPLITUDE OF ACCOMMODATION-PUSH-UP TEST USING RAF RULE

- The patient should wear his or her distance prescription.
- The LE should be occluded.
- The RAF rule is held firmly against the patient's cheeks and tilted inferiorly from the patient's eye level at about 30°.
- The patient is directed to a small reading print on the sliding target, which should be placed at the far end of the rule.
- The procedure involves gradually moving the target closer to the patient.
 - This movement is paused as soon as the patient indicates that the image appears blurry.
 - Then, the patient is instructed to blink several times to attempt to clear their vision.
 - If the print becomes clear after blinking, the target is brought even nearer until the patient again reports a blurred image.
- The test ends when the patient experiences sustained blur, meaning the target becomes blurry and stays that way even after blinking and trying to focus. The position of the target in centimeters is then converted to a dioptric value.
- The procedure is repeated 3 times to obtain an average measurement of the amplitude.

- The procedure is then repeated for the LE with the occluder over the RE.
- The occluder is then removed and the measurement is taken for both eyes. This measurement should usually be greater than the individual 2 measurements provided that there are no binocular vision anomalies present.
- Take note of the distance at which sustained blur is obtained and convert this distance to a dioptric equivalent to reflect the amplitude of accommodation.
- Record properly.

3.1.5 PUPIL EXAMINATION

- Explain the purpose of the test to the patient.
- Provide proper instructions and appropriate fixation target
- Properly measure and state the following:
 - Pupil size OD in bright light
 - Pupil size OS in bright light
 - Pupil size OD in dim light
 - Pupil size OS in dim light
- Properly perform the direct pupillary light response OD and OS
- Accurately state the direct pupillary light responses OD and OS
- Properly perform the consensual pupillary light response OD and OS
- Accurately state the consensual pupillary light responses OD and OS
- Properly perform the swinging flashlight test
- Accurately state the obtained swinging flashlight test findings
- Properly evaluate the near pupillary response
- Accurately state the obtained near pupillary responses OD and OS
- Perform the procedure in a smooth and logical sequence
- Record correctly using standard notation

3.1.6 TONOMETRY-GOLDMANN TONOMETRY

- Explain the purpose of the procedure to the patient
- Position the tonometer probe (tip well aligned with markings at 0/180, drum set at 10 mmHg)
- Instill topical anesthetic/fluorescein safely
- Evaluate the cornea for staining (10X, cobalt filter, max intensity)
- Position the slit lamp for tonometry such that the tip is well illuminated (diffuse, cobalt blue filter, 10-16X, high intensity, wide angle)
- Instruct patient on fixation and hold eyelids properly
- Applanate the cornea safely and efficiently using the appropriate lid holding technique (should not let go of slit lamp throughout applanation)

- Center the mires in view
- Adjust the mires so the inner edge of the rings meets (correct width of mires)
- Read measurement and record correctly (time)
- Examine the cornea for abrasion (10X, cobalt filter, max intensity)
- Switch off equipment
- Conduct the procedure in a smooth and efficient manner

3.1.7 SLIT-LAMP EXAMINATION

- Wash hands and disinfect equipment
- Explain the purpose of the procedure to the patient
- Prepare the biomicroscope correctly (focus eyepieces, allow the examiner to sync view)
- Position the patient properly in the biomicroscope (head position, chin, forehead) throughout the procedure
- Give the patient appropriate fixation targets throughout the test
- Examine and accurately describe the lids, lid margin, and lashes (low light diffuse illumination; wide slit, 6-10X, 30-45 degrees)
- Examine and accurately describe the bulbar conjunctiva (diffuse illumination as above)
- Examine and accurately describe the entire cornea with a parallelepiped. (max illumination, 16X, 30-45 degrees, 1-2 mm) the note requires inferior and superior scan)
- Examine and accurately describe the iris (wide parallelepiped, as above)
- Examine and accurately describe the entire cornea in the optic section. (max illumination, < 0.25 mm, 16X, 30 -45 degrees) note; requires an inferior and superior scan
- Examine and accurately describe the entire cornea using sclerotic scatter. (high-intensity paralleled-piped at the limbal margin)
- Examine and accurately describe the cornea in specular reflection. (angle of incidence=angle of reflection)
- Properly assess the temporal and nasal angles angle with the Van Herrick technique and accurately grade (60 degrees, optic section, max illumination)
- Accurately assess the anterior chamber ((darkroom illumination, high mag, short parallelepiped or conic section 2-3mm spot, 40-50 degrees)
- Examine and accurately describe the lens (narrow parallelepiped or optic section as above)
- Examine and accurately describe the inferior and superior palpebral conjunctiva using diffuse illumination as indicated above (requires lid eversion)
- Conduct the procedure in a smooth, logical sequence (assessing from anterior to posterior structures ending with lid eversion)
- Record correctly using standard notation

3.1.8 FUNDOSCOPY

DIRECT OPHTHALMOSCOPY

- Introduce oneself and extend a welcoming greeting to the patient.
- Wash hands
- Disinfect all equipment before using
- Explain the procedure to the patient.
- Hold the O-scope handle in the right hand for the right eye and vice versa
- Position the o-scope 10 -15 cm from the patient's face, using a spot beam with a +8 to +10D lens focus on the iris, assess and explain the optical clarity of the media by moving about 300 in each direction
- Reduce the plus power of the lens while moving towards the patient to observe the fundus. Assess and describe the retina: Optic disc margin, rim tissue, cup size, and depth. Determine the C/D ratio
- Evaluate and describe the region adjacent to the disc, mid-periphery retinal quadrants, and macula systematically
- Record appropriately

BINOCULAR INDIRECT FUNDOSCOPY WITH 90D, 78D, or 66 D LENSES

- Introduce oneself and extend a welcoming greeting to the patient.
- Wash hands
- Disinfect all equipment before using
- Explain the purpose of the procedure
- Focus slit-lamp eyepieces correctly (to sync view)
- Position the patient properly in the biomicroscope (head position, chin, forehead) throughout the procedure
- Examine and accurately describe the lens (narrow parallelepiped, 10-16X)
- Examine and accurately describe the anterior vitreous
- Using the non-contact fundus lens:
- Properly set up the slit lamp and position lens (10-16X, short parallelepiped, zero degrees)
- Examine and describe the optic nerve (color, rim, elevation)
- Accurately state the vertical C/D ratio
- Accurately state the horizontal C/D ratio
- Examine and accurately describe the vascular arcades
- Examine and accurately describe the fovea and macula
- Record correctly using standard notation
- Conduct the procedure in a smooth and efficient manner
- Properly instruct the patient throughout the procedure

BINOCULAR INDIRECT OPHTHALMOSCOPY WITH 20 D LENS

- Explain the purpose of the procedure

- Focus and center eyepieces correctly
- Adjust the light level properly
- Position the condensing lens correctly
- Obtain a clear image, filling the condensing lens, accurately describing any remarkable pathology of:
 - Superior temporal peripheral retina
 - Superior nasal peripheral retina
 - Nasal peripheral retina
 - Temporal peripheral retina
 - Inferior temporal peripheral retina
 - Inferior nasal peripheral retina
- Obtain a clear image, filling the condensing lens, and accurately describe the appearance (optic nerve, macula, vessels) of the posterior pole
- Record correctly using standard notation
- Properly instruct the patient throughout
- Conduct the exam in a smooth and efficient manner
- Switch off equipment

3.1.9 OBJECTIVE REFRACTION

- Wash hands and disinfect equipment.
- Set the patients interpupillary distance in the trial frame.
- Place the trial frame comfortably on the patient with the patient's face well centered (ensure 3-point touch).
- Tell the patient to keep both eyes open and inform you when you block them.
- Keep your one eye open.
- Use the right eye to examine the right eye and the left eye for the left eye.
- Hold the retinoscope at your normal working distance 50 or 67cm.
- Insert your working distance lens 2D or 1.5D.
- Hold the retinoscope in your right hand while examining the right eye and vice versa.
- Reduce room illumination.
- Ask the patient to look at the 6/60 target.
- Determine if the refractive error is spherical or astigmatic by changing the sleeve of the ret or changing position for a clear streak.
- Check the thickness of the reflex.
- Check for the break phenomenon.
- Identify (and state) with or against motion.
- Neutralize using the appropriate lens (plus for with motion and minus for against motion).

- If astigmatism is present, identify the principal meridians.
- Neutralize the least myopic or most hyperopic first.
- Use the right cylinder power to neutralize the other meridian with the axis rightly aligned.
- Place the sphere in the lens well closest to the patient and the cylinder towards the front of the trial frame.
- Recheck neutralization in all meridians and in the plane and concave mirror.
- Remove your working distance lens.
- Recheck VA through the net Retinoscopic findings (OD / OS).
- Record net Retinoscopic findings in minus cylinder for.

3.10 SUBJECTIVE REFRACTION

Best Vision Sphere

- Explain to the patient the purpose of the procedure
- Wash hands and disinfect equipment.
- With net Retinoscopy findings in place, Occlude the left eye.
- Check VA
- Start with the plus probe lens (+0.25 / +1.00) depending on VA
- Direct the patient's attention to a particular letter on the line above the best line VA with net ret finding.
- Ask the patient if the letter is brighter, sharper and clear with or without the probe lenses.
- If the patient says the addition of the +0.25 is better, keep adding lenses in +0.25d steps until the letter is blurred, if the patient rejects +0.25 try -0.25DS if he prefers it, keep adding -0.25DS until VA improves.
- Stop if the letters are getting smaller.
- Leave the eye with either maximum plus or minimum minus that provides the best VA.

Astigmatic Error Determination OD.

- Check VA and select a circular target one line above the best VA.
- To refine the axis, place the handle of the JCC to axis line of the trial cylinder lens axis
 - flip the JCC and ask the patient which position makes the target appear rounder, sharper and clearer
 - based on the patient choice move the net ret cylinder axis in the direction of the JCC minus power line (red).
- The first movement of the trial cylinder axis should be by 15 or 20 degrees and the second movement of the trial cylinder axis can be by 5 or 10 degrees depending on patient responses.
 - The end point of axis refinement is when the patient says the two positions are the same. Alternatively, choose the middle.

- To refine power, place the JCC with the power lines parallel to the axis of the trial cylinder then flip the JCC and ask which position makes the target sharper, clearer and darker.
 - if patient chooses the +0.25 side of the JCC patient has no astigmatism
- If the patient chooses the -0.25 side of the JCC, increase the cylinder power by 0.25 steps as you flip and repeat the question in step above.

NB: For each -0.50 Cylinder power change, add +0.25 lens.

- The end point is when the patient chooses +0.25 side or the two sides are the same.
- Remeasure VA and record properly.

Monocular Balancing OD

- Explain the procedure to the patient clearly and give proper instruction.
- Turn room lights off.
- Direct the patient's attention to the duochrome test targets.
- Ask which number or target appears darker sharper and clearer between the two backgrounds and If the target on the red background appears darker, sharper and clearer add -0.25 until both backgrounds are clearly equal
- If the target on the green background is better add +0.25 until the targets on both backgrounds are clearly equal or targets on the green are slightly better.
- Record the subjective findings and monocular acuity (OD/OS)

4.0 Treatment Protocols for Frequent Conditions

4.1 Chalazion Management

- Natural Resolution: Approximately one-third of chalazions resolve without treatment.
- Warm Compresses: Apply warm compresses to the affected area for 10 minutes four times daily (q.i.d.), accompanied by gentle massage over the lesion.
- Topical Antibiotics: Use antibiotic ointments such as Bacitracin or Erythromycin twice daily (b.i.d.).

For Persistent Chalazions:

- Surgical Approach: Perform an incision and curettage of the cyst.
- Steroid Injections: Administer Triamcinolone injections, especially if the chalazion is close to the lacrimal apparatus.
- Systemic Tetracycline: Use as a preventive measure in cases of recurrent chalazion.

4.2 External Hordeolum (Stye)

- Apply warm compresses to the affected area for 10 minutes, four times daily (q.i.d.), with gentle massage over the lesion.

- Use a topical antibiotic such as Bacitracin or Erythromycin ointment twice daily (b.i.d.), or alternatively,
- Chloramphenicol or Ciprofloxacin eye drops four times daily (q.i.d.).
- Perform epilation of the eyelashes if necessary.
- If the condition persists, refer the patient for incision and drainage.

4.3 Bacterial conjunctivitis

- Approximately 60% of cases improve without any treatment.
- Antibiotics are often used to hasten recovery and prevent re-infection.
- In adults, a broad-spectrum antibiotic eye drop, such as Chloramphenicol, Ciprofloxacin, Ofloxacin, Gentamicin, or Polymyxin B, is recommended.
- These should be administered every 2 hours during waking hours for a period of 5-7 days.
- For children, compliance is generally higher when using gel formulations or antibiotics like Chloramphenicol, Gentamicin, or Tetracycline.

4.4 Gonococcal conjunctivitis

- Topical Treatment:
- Administer Gentamicin or Bacitracin topically, starting with an hourly application.
- Systemic Treatment:
- Refer for administration of Ceftriaxone intramuscularly, with a dosage of 250 mg daily for 3 days, or initiate treatment with a 1 g dose.
- For patients with Keratitis, a more aggressive treatment approach is required, involving up to 2g administered intravenously for 3 days.
- Additionally, emphasize the importance of maintaining good facial hygiene and to get their partners to receive the same treatment.

4.5 Viral conjunctivitis

- Artificial tears 4 to 8 times per day for 1 to 3 weeks
- Cool compresses several times per day for 1 to 2 weeks
- Vasoconstrictor/Antihistamine e.g. Naphazoline/ Pheniramine q.i.d. if itching is severe
- If a membrane/ pseudo-membrane is present it is gently peeled.
- In severe cases; Topical Steroids e.g. Fluorometholone 0.1% or Prednisolone acetate 0.125% q.i.d. for a week then tapered.
- Counsel the patient- Self-limiting condition, gets worse 4 to 7 days and may not resolve for 2 to 3 weeks.
- Very Contagious; advise on precaution not to spread.

- Frequent Hand washing.

4.6 Allergic conjunctivitis

- Remove the causing agent.
- Apply cool compresses to the affected area multiple times a day.
- Use topical eye drops based on the severity of the condition:
- Mild Cases:
 - Artificial tears, applied 4 to 8 times daily.
- Moderate Cases:
 - Vasoconstrictor/Antihistamine, four times daily (q.i.d.), such as Naphazoline/Pheniramine. Note the risk of rebound vasodilation with extended use.
 - Ketotifen 0.025%, twice daily (b.i.d.), to alleviate itching.
 - Ketorolac 0.5%, four times daily (q.i.d.), to lessen symptoms.
 - Mast Cell Stabilizers, like Sodium Cromoglycate, four times daily (q.i.d.).
- Severe Cases:

A mild topical steroid, for example, Fluorometholone 0.1%, four times daily (q.i.d.) for 1 to 2 weeks.
- Additionally, consider oral antihistamines such as Fexofenadine 60 mg orally, twice daily (b.i.d.); Loratadine 10 mg orally, once daily (q.d.); or Cetirizine 5 to 10 mg orally, once daily (q.d.).

4.7 Blepharitis

Lid Hygiene Protocol:

1. Warm Compress:

- Apply a warm compress for several minutes to loosen crusts at the bases of the eyelashes.

2. Lid Cleaning:

- Clean the eyelids using baby shampoo or a solution of sodium bicarbonate.

3. Antibiotics:

- Topically: Use Bacitracin or Chloramphenicol eye ointment or drops.
- Orally: Administer Azithromycin 500 mg daily for 3 days.

4. Weak Topical Steroid:

- Fluorometholone 0.1%, applied four times daily (q.i.d.) for one week, particularly in patients with severe papillary conjunctivitis, marginal keratitis, or Phlyctenular conjunctivitis.

5. Tear Substitutes:

- Administer as needed to provide relief.

4.8 Pinguecula

- Treatment is typically not required; however, tear substitutes may offer relief.
- In cases where the Pinguecula is inflamed, a short course of Fluorometholone can be beneficial.
- It is also advisable to avoid exposure to dust and direct sunlight.

4.9 Pterygium

Medical Management:

- Utilize tear substitutes for lubrication and comfort.
- Apply topical steroids to alleviate inflammation.
- Recommend wearing sunglasses to minimize UV exposure and reduce factors that stimulate growth.

Surgical Intervention:

- Consider simple excision in cases where there's high recurrence or if the Pinguecula obstructs the visual axis.
 - However, this approach is associated with a high rate of recurrence.
- Techniques such as conjunctival autograft or the use of amniotic membrane are widely employed for more effective results.

4.10 Dry eye

- Patient Education:

Setting Expectations and Compliance:

- Clearly explain the expected outcomes and stress the importance of adhering to treatment plans.

Avoidance of Harmful Substances:

- Educate on avoiding toxic drugs or environmental factors that may worsen the condition.

Encouraging Healthy Habits:

- Emphasize the need to blink regularly while reading or using visual display units (VDUs).

Addressing Secondary Causes:

- Manage underlying causes such as Rheumatoid arthritis or issues related to contact lens wear.

Protective Measures:

- Advise wearing sunglasses to shield eyes from the sun.
- Suggest the use of tear substitutes, including drops and gels:
- Cellulose derivatives like Hydroxymethyl cellulose for mild cases.
- Polyvinyl alcohol to improve tear film persistence, beneficial for mucin deficiency.
- Ointments such as Petrolatum mineral oil, especially for use at bedtime.

Mucolytic Agents:

- Acetylcysteine 5% drops four times daily (q.i.d.) can be effective for patients with corneal filaments.

Punctal Occlusion:

- Consider temporary, reversible, or permanent options, particularly for patients with moderate to severe Keratoconjunctivitis Sicca (KCS) or any degree of eye dryness.

Anti-inflammatory Treatments:

- Low dose topical steroids, though their long-term use is controversial.
- Topical Cyclosporine to reduce T-cell mediated inflammation.
- Systemic Tetracyclines as an option.

Management Related to Contact Lenses:

- Focus on conserving existing tears and reducing room temperature to alleviate discomfort.

Additional Options:

- Explore other treatments like Tarsorrhaphy if needed.

4.11 Trachoma-SAFE.

Surgical Intervention:

- Tarsal Plate Rotation (TPR) to correct conditions such as entropion or ectropion.

Antibiotic Treatment:

- A single dose of Azithromycin, 20 mg/kg (maximum of 1 g), or

- Doxycycline, 100 mg orally, twice daily (b.i.d.), or
- Tetracycline, 250 mg orally, four times daily (q.i.d.) for 2 weeks.
- For women of childbearing age, Erythromycin 500 mg twice daily (b.d.) for 14 days is an alternative.
- Note: Topical 1% Tetracycline ointment is less effective than oral treatments.

Facial Hygiene:

- Emphasize the importance of regular face washing and controlling fly exposure to reduce infection risks.

Environmental and Sanitation Measures:

- Ensure access to clean water and improvement of general sanitation conditions.

4.12 Corneal ulcer

Bacterial Keratitis Treatment

Topical Antibiotics:

- Fortified antibiotics administered hourly for the initial 24-48 hours, day and night.
- Frequency reduced to every 2 hours during waking hours for the next 48 hours, then four times daily (q.i.d.) for a week.

Preparation examples:

- Gentamicin or Tobramycin 15mg/ml: Dilute 2 ml of 40mg/ml parenteral antibiotic with 4 ml of 0.3% ophthalmic solution.
- Cefazolin 50 mg/ml: Mix 500 mg of parenteral antibiotic with 2.5 ml sterile water, then add to 7.5 ml of preservative-free artificial tears. This mixture remains stable for 24 hours at room temperature or 96 hours refrigerated.

Oral Antibiotics:

- Ciprofloxacin 750 mg twice daily for 7-10 days, typically reserved for severe cases like threatened or actual corneal perforation.

Subconjunctival Antibiotics:

- Consider if there is poor compliance with topical treatment.

Mydriatics:

- Use Atropine 1% or Cyclopentolate 1% to prevent posterior synechiae and alleviate pain from ciliary spasm.

Pain Management:

- Oral painkillers as needed.

Topical Steroids:

- Used only if bacterial infection is unconfirmed.

Hospital Admission:

- Recommended for sight-threatening infections.

Fungal Keratitis Treatment

Epithelium Removal:

- Enhances antifungal agent penetration.

Topical Treatment:

- Intensively applied hourly for the first 48 hours, then reduced as symptoms permit.

Duration:

- Continuation for several weeks as most antifungals are fungistatic.

Specific Treatments:

- Filamentous infections: Natamycin 5% or Econazole 1%; alternatives include Amphotericin B 0.15% and Miconazole 1%.
- Candida infections: Usually Econazole 1%, with alternatives like Natamycin 5%, Fluconazole 2%, Amphotericin B 0.15%, and Clotrimazole 1%.

Cycloplegics:

- Example: Scopolamine 0.25% thrice daily.

Topical Steroids:

- Avoid or taper rapidly if used.

Severe Cases:

- Subconjunctival Fluconazole for severe cases with hypopyon.
- Systemic antifungals like Itraconazole 100mg daily or Voriconazole 100 mg with a 200 mg loading dose.
- Excisional Penetrating Keratoplasty may be needed for unresponsive cases.

Viral Keratitis Treatment

Herpes Simplex Keratitis:

- Topical antivirals like Triflurothymidine or Acyclovir.
- Acyclovir 3% ointment applied five times daily.
- Debridement for dendritic, not geographic, ulcers.
- Warm or cool soaks to skin lesions thrice daily.

Herpes Zoster Ophthalmicus:

- Oral Acyclovir 800 mg five times daily for 3-7 days, started within 72 hours of onset.
- Treatment is still beneficial after 72 hours for new vesicles to reduce severity and risk of post-herpetic neuralgia.
- Topical Cycloplegic (e.g., Scopolamine 0.25% t.i.d.) for intraocular inflammation.
- Intravenous Acyclovir 5-10 mg/kg thrice daily indicated for encephalitis.
- Systemic Steroids (e.g., Prednisolone 40-60 mg daily) used with systemic antivirals.
- Symptomatic treatment of skin lesions includes drying, antisepsis, and cold compresses.

4.13 Glaucoma

Acute Angle Closure Treatment

Immediate Actions:

- Administer Carbonic Anhydrase Inhibitors such as Acetazolamide 500 mg intravenously, followed by 500 mg orally, unless the patient is vomiting.

Topical Medication:

- Use topical beta-blockers, like Timolol 0.5%, in a single dose, unless contraindicated.
- Provide Analgesia and Antiemetics as needed.

One Hour Later:

- Administer Pilocarpine 2% four times daily (q.i.d.).
- Prophylactically treat the fellow eye with Pilocarpine 1% q.i.d. until a preventive laser iridectomy is performed.

Thirty Minutes Later:

- If intraocular pressure (IOP) remains above 35 mmHg, administer oral 50% Glycerol 1g/kg, with caution in diabetic patients.
- In cases of vomiting or inability to tolerate oral Glycerol, give 20% Mannitol (1-2 g/kg) intravenously over 45 minutes.
- Nd: YAG Laser Iridectomy as needed.
- Note: Glycerol and Mannitol function as osmotic diuretics.

Primary Open Angle Glaucoma Treatment

Beta Blockers:

- E.g., Timolol, reduces aqueous production, typically administered twice daily.
- Caution advised in asthma or heart disease patients.

Prostaglandin Agonists:

- E.g., Latanoprost (Xalatan or Lumigan) or Travaprost, increases drainage, usually given once at night.
- Often used in combination with beta blockers, such as in Xalacom (Timolol and Latanoprost).

Selective Alpha 2 Receptor Agonists:

- E.g., Brimonidine (often branded as Alphagan), caution advised due to potential hypertension, especially in patients on monoamine oxidase inhibitors.

Miotics:

- E.g., Pilocarpine, facilitates drainage by pulling the iris away from the anterior angle.

Carbonic Anhydrase Inhibitors:

- Systemic: E.g., Acetazolamide.
- Topical: E.g., Dorzolamide.

Surgical Options:

- Often preferred due to issues with eye drop compliance.
- Surgery, like Trabeculoplasty, is effective but requires ongoing IOP monitoring. Surgery may need to be repeated or supplemented with eye drops after about 5 years.

4.14 Acute Iritis

Mydriatics

Short-Acting Mydriatics:

- Tropicamide (0.5% and 1%): Effective for about 6 hours.
- Cyclopentolate (0.5% and 1.5%): Lasts for approximately 24 hours.
- Phenylephrine (2.5% and 10%): Active for around 3 hours but lacks cycloplegic effects.

Long-Acting Mydriatics:

- Homatropine 2%: Effects last up to 2 days.

- Atropine 1%: The most potent mydriatic and cycloplegic, with effects lasting up to 2 weeks.

Indications:

- Used to provide comfort.
- Helpful in breaking down recently formed posterior synechiae.

Topical Steroids:

- Start with intensive therapy, applying hourly or every minute for the first 5 minutes of every hour.
- Once inflammation is under control, taper to every 2 hours, then every 3 hours, and eventually four times daily, reducing by one drop each week.
- Typically discontinued by 5-6 weeks.

Severe Cases:

- Periocular Steroids, Intraocular Steroids may be necessary.

Systemic Steroids:

- Oral administration (Prednisolone 5mg or 25 mg), or
- Intravenous injection (Methylprednisolone 1g/day for 2-3 days).
- Address secondary causes, such as Glaucoma or Juvenile Rheumatoid Arthritis (JRA).

4.15 **Orbital Cellulitis**

Hospitalization with regular evaluations by both ophthalmology and otolaryngology specialists is essential. The treatment plan includes:

Antimicrobial Therapy:

- Intramuscular Ceftazidime 1 g administered every 8 hours, or
- Intravenous Ampicillin 1.5 to 3 g every 6 hours (q6h), and
- Oral Metronidazole 500 mg every 8 hours for anaerobic coverage.

For Patients with Penicillin Allergy:

- Intravenous Vancomycin as an effective alternative.

Monitoring of Optic Nerve Function:

- Conduct assessments every 4 hours, including tests for pupillary reactions, visual acuity, color vision, and light brightness perception.

Investigative Procedures:

- Blood examinations -Complete Blood Count (CBC), Erythrocyte Sedimentation Rate (ESR) and C-Reactive Protein (CRP), Blood Cultures
- CT scans of the orbit,
- Lumbar puncture if there are indications of meningeal or cerebral involvement.

Surgical Intervention:

- Considered for cases that do not respond to medical treatment.

4.16 Headache

To address migraines, consider the following approach:

Identifying and Managing Underlying Triggers:

- Avoid known headache triggers such as alcohol, certain foods, stress, loud noise, or bright light.

Vision Correction:

- Correct any significant refractive errors which may contribute to headaches.

3. Medication Strategy:

- Initial Therapy: Use Aspirin or nonsteroidal anti-inflammatory agents like Tolfenamic acid 200 mg every 3 hours.
- More Potent Options (if initial therapy is ineffective): Ergotamine 2 mg as a single dose, then repeat every 30 minutes for two additional doses if necessary (maximum of 6 mg per attack).
 - Combination of Ergotamine 1 mg with Caffeine 100 mg, two tablets at onset followed by one tablet every 30 minutes.
 - Dihydroergotamine, 4.0 mg orally as a single dose.

Note: Ergotamine is not recommended for elderly patients, or those with cardiovascular, renal, hepatic conditions, or during pregnancy.

Prophylactic Medication (for frequent or severe headache attacks):

- Propranolol 10 to 80 mg orally, daily in divided doses, gradually increasing by 10-20 mg every 2 to 3 days until the desired effect is achieved.
- Amitriptyline: 25 to 200 mg orally at bedtime.
- Calcium Channel Blocker, such as Lunarizine, 5 to 10 mg daily.

Antinausea Treatment (for acute episodes):

- Prochlorperazine 25 mg administered rectally, twice daily.

These recommendations aim to both treat acute migraine attacks and reduce their frequency and severity.

4.17 Senile Cataract

Indications for Cataract Surgery:

Visual Improvement:

- The primary reason for surgery, varying individually. Surgery is considered when the cataract progresses to a point where it interferes with daily essential activities.

Medical Reasons:

- Situations where the cataract negatively impacts the health of the eye, such as in cases of phacolytic glaucoma or diabetic retinopathy.

Cosmetic Reasons:

- Rarely, surgery is performed to restore the normal appearance of the pupil, changing it from white back to black.

Preoperative Evaluation:

Visual Acuity Assessment:

- To determine the current level of vision.

Cover Test:

- Used to detect Heterotropia, which may indicate Amblyopia, influencing the prognosis.

Pupillary Responses:

- Checking for afferent pupillary defects which may suggest other ocular pathologies.

Ocular Adnexa Examination:

- Identifying issues like Dacryocystitis, Blepharitis, Ectropion, or chronic conjunctivitis that may increase the risk of Endophthalmitis.

Cornea Evaluation:

- Checking for stromal opacities and endothelial cell count, which affect surgical outcomes.

- Measuring corneal astigmatism.

Anterior Segment Inspection:

- Assessing anterior chamber depth and the presence of pseudoexfoliation, which could indicate weak zonules and potential surgical complications.

Lens Examination:

- Identifying the type of cataract, as nuclear cataracts are typically harder than cortical ones.

Fundus Examination:

- Assessing any underlying pathology such as Age-Related Macular Degeneration (AMD) that could affect visual outcomes.

Biometry:

- Measuring intraocular lens power for implantation after cataract surgery.

Other Measurements:

- A-Scan for axial length, Keratometry for corneal power, and determining the appropriate constant and intraocular lens power using formulas like the SRK (Sanders, Retzlaff, Kraff) Formula.

Types of Cataract Surgery:

Intracapsular Cataract Extraction (ICCE):

- With anterior chamber intraocular lens (ACIOL).

Extracapsular Cataract Extraction (ECCE):

- With posterior chamber intraocular lens (PCIOL).

Phacoemulsification:

- With foldable PC IOL.

Small Incision Manual Cataract Surgery (SICS):

- With PC IOL.

Prophylactic Treatment:

UV Protection:

- Using sunglasses and peak caps.

Vitamin Supplementation:

- Including Vitamin C, Vitamin E, Beta carotene, zinc oxide, and copper oxide..

4.18 Treatment of Wet ARMD

Treatment modalities for certain retinal conditions can include:

Intravitreal Steroids:

- Specifically, Triamcinolone acetonide is used for its anti-inflammatory properties.

Anti-Vascular Endothelial Growth Factor (Anti-VEGF) Therapy:

- An example is Intravitreal bevacizumab (Avastin), which is employed to inhibit abnormal blood vessel growth and leakage.

Laser Photocoagulation:

- This technique uses focused laser light to treat specific areas of the retina, useful in conditions like diabetic retinopathy.

Surgery: Sub-macular like vitrectomy, posterior retinotomy

Low vision aids

4.19 Treatment of Retinal Vein Occlusion

For the management of certain ocular conditions, the following medical investigations and treatments are recommended:

Medical Investigations:

- Blood Pressure Measurement: To assess hypertension, which can impact ocular health.
 - Electrocardiogram (ECG): To evaluate heart function, as systemic heart conditions can affect the eyes.
 - Blood Tests: To check for underlying conditions like diabetes or high cholesterol that can influence eye health.
 - Chest X-Ray: To rule out any respiratory or cardiovascular issues that may have ocular implications.

4.20 Treatment for Macular Edema

- Grid Laser Photocoagulation: A laser treatment that targets specific areas of the retina to reduce swelling and improve vision.

- Intravitreal Triamcinolone Acetonide: An injection into the eye of a steroid medication to reduce inflammation and swelling in the macula.

4.21 Treatment of Retinal Artery Occlusion

The following procedures are utilized in specific ocular emergencies or conditions to manage intraocular pressure (IOP) and improve ocular blood flow:

Ocular Massage:

- This technique involves the use of a three-mirror contact lens. It's applied for approximately 10 seconds to achieve central artery pulsation or to momentarily stop blood flow, followed by a 5-second release period. The objective is to mechanically compress the arterial lumen, inducing prompt changes in arterial blood flow.

Anterior Chamber Paracentesis:

- This procedure involves the removal of a small amount of fluid from the anterior chamber of the eye. It's performed to rapidly reduce IOP.

Intravenous Acetazolamide:

- Administering Acetazolamide intravenously is aimed at achieving a more sustained reduction in IOP than what might be accomplished with repeated paracentesis alone.

These interventions are often critical in managing conditions that involve elevated intraocular pressure or compromised ocular blood flow.

4.22 Treatment of retinitis Pigmentosa

The recommended management for certain nutritional and ocular conditions includes dietary modifications and supplementation as follows:

Vitamin A Supplementation:

- Water-miscible vitamin A, ranging from 10,000 to 15,000 International Units (IU), to be taken orally once daily (q.d.).

Vitamin E Supplementation:

- Administer 200 to 300 IU/kg of Vitamin E orally once daily (q.d.).

Vitamin K Supplementation:

- Vitamin K at a dose of 5 mg, to be taken orally once weekly.

Dietary Fat Restriction:

- Limit dietary fat to 15% of total caloric intake to help manage or prevent certain conditions.

Regular Monitoring:

- Conduct biannual blood tests to monitor serum levels of vitamins A and E.
- Perform yearly electroretinography (ERG) and dark adaptometry to assess retinal function.

Zinc Supplementation:

- Consider adding zinc to the patient's diet, as it can be beneficial for eye health.

Low Vision Aids:

- Recommend or provide low vision aids to assist in daily activities, especially in cases of impaired vision.

These interventions are designed to address specific nutritional deficiencies and their impact on ocular health while also providing support for low vision.

4.23 Treatment of Papilledema.

For managing conditions associated with raised intracranial pressure, the following treatment protocol is typically recommended:

Discontinue Causative Drugs:

- Stop any medications implicated in causing increased intracranial pressure.

Regular Visual Field Testing:

- Conduct regular perimetry tests to detect early or progressive changes in the visual field.

Diuretic Medication:

- Use diuretics like Acetazolamide or Thiazides as a first-line treatment for headache relief.

Weight Management:

- Encourage weight loss in obese patients, as this can help reduce intracranial pressure.

Optic Nerve Fenestration:

- This surgical procedure involves making an incision in the meningeal sheath surrounding the optic nerve.
- It is considered safe and effective for preserving vision, particularly when performed early in the course of the disease.

4.24 Treatment of Hypertensive Retinopathy

- Control of Hypertension: Key in managing retinopathy.

Based on Grades:

- Grade 0: No retinal changes.
- Grade 1: Mild arterial changes, vein concealment.
- Grade 2: Obvious arterial changes, vein deflection at crossings.
- Grade 3: Includes copper-wiring of arteries, venous changes, hemorrhages, and/or exudates.
- Grade 4: All above plus silver-wiring of arteries and disc swelling.

4.25 Treatment of Diabetic Retinopathy

Control of Diabetes: Crucial for management.

- For Macular Edema: Focal/grid laser for clinically significant cases.

Proliferative Diabetic Retinopathy: Pan-retinal photocoagulation.

For Vitreous Hemorrhage: Vitrectomy.

4.26 Xerophthalmia

WHO Grading:

- XN: Night blindness.
- X1: Conjunctival xerosis; Bitot's spots.
- X2: Corneal xerosis.
- X3: Corneal ulceration.
- XS: Corneal Scar.
- XF: Xerophthalmic fundus.

Treatment:

- Dietary: Vitamin A-rich fruits, green leafy vegetables.
- Vitamin A Therapy: Dosage varies with age and condition.

- Intensive ocular lubrication; treat malnutrition.
- Zinc supplementation.
- Keratoplasty for corneal scars.

4.27 Eye Injuries

Chemical Injuries:

- Immediate Treatment: Extensive irrigation, lid eversion, debridement, IOP monitoring.
- Grading:
 - Grade 1: Clear cornea, no ischemia.
 - Grade 2: Hazy cornea, minor ischemia.
 - Grade 3: Total epithelium loss, significant ischemia.
 - Grade 4: Opaque cornea, major ischemia.

Treatment:

- Mild (Grade 1 and 2): Topical steroids, cycloplegics, antibiotics.
- Severe Burns: Anti-inflammatory measures, promote epithelial regeneration, prevent ulceration. Use cycloplegics, antibiotics, steroids, antiglaucoma drugs, conjunctival adhesions lysis, artificial tears, ascorbic acid, sodium citrate, and tetracyclines. Consider therapeutic contact lenses.

Conclusion:

As we conclude this manual, we hope that it has provided you with a comprehensive understanding of the complex world of eye care. The field of optometry is ever-evolving, and it is our responsibility as healthcare professionals to stay informed and adaptable. We encourage you to use this manual not just as a reference, but as a guide in your continued professional development. Remember, the ultimate goal of our profession is to enhance the quality of life of our patients through exceptional eye care. We trust that this manual will be a valuable asset in your endeavor to achieve this noble objective. Keep learning, keep growing, and continue to make a difference in the lives of those you serve.

Appendix

Clinical evaluation form

MZUZU UNIVERSITY
FACULTY OF HEALTH SCIENCES
DEPARTMENT OF OPTOMETRY

Clinical evaluation form

Name of student..... Date.....

Reg. number.....

CASE HISTORY	EXPECTATIONS		Comments
	Chief Complaint (<i>dr. fallopes</i>)	2	
	History of presenting complaint	1	
	Patient social demographics	1	
	Patient ocular History	1	
	Patient general history	1	
	Medications	1	
	Family ocular History	1	
	Family general history	1	
	Allergies	1	
	Good Recording	1	
PRELIMINARY			
Dist. Visual Acuity	Instructions and purpose of the test	1	
	Procedure and recording	2	
Near Visual Acuity	Instructions and purpose of the test	1	
	Procedure and recording	2	
Pinhole	Instructions and purpose of the test	1	
	Procedure and recording	2	
Visual Field (Confrontation etc)	Instructions and purpose of the test	1	
	Procedure and recording	2	
Cover Test (D and N)	Instructions and purpose of the test	1	

	Check for Tropia Correctly	2	
	Check for Phoria Correctly	1	
	Cover/Uncover done Correctly	1	
	Alt. Cover Test is done Correctly	1	
	Recording	1	
Motility	Instructions and purpose of the test	1	
	Procedure and recording	2	
NPC	Instructions and purpose of the test	1	
	Procedure and recording	2	
Pupils	Instructions and purpose of the test	1	
	Procedure: Direct/ Consensual	2	
	Record properly	1	
Amplitude of accommodation	Instructions and purpose of the test	1	
	Procedure	2	
CASE-SPECIFIC DIAGNOSTIC TESTS (IOP, BRUCKNER, VON GRAEFFE, TBUT ETC)			
	Instructions and purpose of the test	1	
	Procedure and recording	2	
REFRACTION			
Retinoscopy	Instructions to Patient	1	
	Give appropriate target/use the appropriate fogging lens	1	
	Procedure	4	
	Record and indicate VA	2	
Subjective	Instructions to Patient	1	
	Use appropriate target	1	
	Best Sphere determination	2	
	Use the JCC technique to determine the axis, Cylinder	3	

	Power, and adjust the sphere component.		
	Binocular balancing	2	
	Record the findings properly	1	
Near Add (3)	Instructions given	1	
	Estimate ADD by Age or tentative formula or calculated from Amp of Accommodation.	2	
	Refine the ADD trial frame	2	
	Record	1	
OCULAR HEALTH EXAMINATION			
Anterior Eye Examination	Instructions given to the patient	1	
	Examine the adnexa using the right procedure	2	
	Examine the cornea using the right procedure	2	
	Examine Iris using the right procedure	2	
	Examine Anterior Chamber and Ant angle (Shadow test/VH)	2	
	Examine lens procedurally	2	
	Carry out the whole procedure using the logical flow	2	
	Record properly	1	
Direct Ophthalmoscopy	Instructions and purpose of the test were explained to the patient	1	
	the disc properly	2	
	Check and describe the periphery retina properly (8 sections).	2	
	Check and describe macular area properly	2	
	Record the findings properly	1	
CASE ANALYSIS			

Differential diagnosis	Is the student able to utilize data from various tests done to come up with the working diagnosis and give reasons?	5	
DIAGNOSIS			
	Correct tentative diagnosis	4	
MANAGEMENT			
	Has the student addressed the chief complaint?	2	
	Has the student given information on what the patient's problem is?	2	
	Has the student counseled the patient appropriately?	2	
	Has the student referred the patient for further management?	2	
	Has the student advised the patient for follow up	2	
TOTAL		/109	%

NB: On case specific diagnostic tests the supervisor will need to ask the student clinician questions as to why he did not do either of the tests in that section. So, marks can still be awarded to the student based on the reasons given by the student.

Supervisors name.....Signature.....

Topic/Case presentation form

MZUZU UNIVERSITY

FACULTY OF HEALTH SCIENCES

DEPARTMENT OF OPTOMETRY

Topic/Case presentation Grading form

Student name.....

Date.....

Reg. number.....

Topic/Case.....

Assessor.....

Signature.....

		Mark	
Non-verbal Skills			
1	Presenter relaxed 25 50 75 100%	4	
2	Presentation flowing without repetition 25 50 75 100%	4	
3	Mannerism 25 50 75 100%	4	
4	Hesitation 25 50 75 100%	4	
Confidence			
4	Presenter displaying signs of tension 25 50 75 100%	4	
5	Presenter having relaxed hand movement 25 50 75 100%	4	
6	Good Stance 25 50 75 100%	2	
7	Good Poise 25 50 75 100%	2	
Connection			
8	Presenter maintain audience contact 25 50 75 100%	4	
9	The presenter depends much on notes 25 50 75 100%	4	
VERBAL			
Clarity			
10	Presenter easily understood 25 50 75 100%	6	
11	Showing connection of points 25 50 75 100%	6	
Volume			
12	The presenter easily heard 25 50 75 100%	4	
13	Fitting Volume 25 50 75 100%	2	
CONTENT	For case, presentation content is assessed based on the footnote		
Info			
14	Material well-elaborated 25 50 75 100	8	
15	Presenter answer the question with further explanation	6	
Organization			
16	Material presented logically 25 50 75 100%	6	

17	Well organized 25 50 75 100%	4	
18	Presented with clear themes/idea 25 50 75 100%	4	
Visual or graphics			
19	PowerPoint fonts easy to read 25 50 75 100%	6	
20	PowerPoint Background with good contrast 25 50 75 100%	4	
21	PowerPoint has relevant visual aids 25 50 75 100%	6	
PROFESSIONALISM			
Dressing			
22	Dressed comfortably	2	
23	Presenter dressed appropriately	2	
Grooming			
24	Grooming fitting and appropriate	2	
Assessor Signature:		100	

Reference-1 Slide

N/B

- Uses only bullets, the slides should not be wordy.
- Limit the bullets to six per slide.
- The presentation should have at least one visual aid.

Footnote:

- case presentation should be organized and presented in this order-patient history/examination findings/Differential diagnosis/Diagnosis/Discussion-(etiology/prevalence/risk factor/pathophysiology/complications/diagnostic workup) Treatment and management
- Topic presentation must have-introduction/etiology/prevalence/risk factor/pathophysiology others can be included depending on the topic